

Skip-gram Word Embeddings with Negative Sampling

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Abstract

This report presents a NumPy implementation of skip-gram word embeddings trained with negative sampling. We train 200-dimensional embeddings on the provided Wikipedia 1B corpus using two context window sizes (5 and 15) and evaluate the resulting vectors using similarity and analogy prompts. Two bonus analyses are included: combining target and context embeddings at evaluation time, and a small gender-bias probe using nearest neighbors and a gendered analogy. Results show that larger context windows increase runtime and shift neighborhoods toward broader topical associations; combining embeddings can yield slightly cleaner neighborhoods; and gendered terms exhibit stereotypical associations inherited from the corpus.

to words that occurred at least five times. The statistics were obtained from the training trace.

Category	Value
Training corpus	wiki-1B.txt
Initial vocabulary size	253,855
Final vocabulary size (min count = 5)	71,291
Retained training tokens	16,718,845
Embedding dimension	200
Negative samples	5
Subsampling rate	10^{-4}
Iterations	3
Context sizes	5, 15

Table 1: Corpus statistics and training hyperparameters.

1 Introduction

Word embeddings map tokens to dense vectors such that semantic and syntactic regularities are reflected by geometric structure. The skip-gram model learns these vectors by predicting nearby context words from a center word. Negative sampling provides an efficient training objective by contrasting observed word–context pairs with sampled negative pairs. In this assignment, we implement missing components of a provided NumPy codebase and evaluate learned embeddings under two context window sizes. We also report two bonus experiments: combined embeddings and a small bias probe.

2 Model and Training Setup

2.1 Objective

For a target word w and context word c , skip-gram with negative sampling maximizes:

$$\log \sigma(\mathbf{v}_w^\top \mathbf{u}_c) + \sum_{i=1}^K \log \sigma(-\mathbf{v}_w^\top \mathbf{u}_{n_i}),$$

where $\mathbf{v}_w$ is the target embedding, $\mathbf{u}_c$ is the context embedding, $\{n_i\}$ are negative samples, and K is the number of negatives.

2.2 Training Data and Hyperparameters

Training was performed on the `wiki-1B.txt` corpus. The vocabulary was learned directly from the corpus and restricted

2.3 Training Time

Table 2 summarizes wall-clock durations computed from start/end timestamps in `traces.txt`. The larger context increases runtime because more context pairs are generated per target word.

Table 2: Training time by context window size (from `traces.txt`)

Context	Start (Feb 16, 2026)	Duration
5	17:48	1h 08m 51s
15	18:57	2h 57m 03s

3 Core Experiments

We evaluated the trained embeddings using the provided evaluation script on similarity and analogy prompts.

3.1 Word Similarity

Tables 3 and 4 show representative top neighbors (cosine similarity) for the required similarity queries.

Table 3: Top neighbors (context = 5)

Query	Top Neighbors (examples)
book	books, memoir, afterword, preface, autobiographies
trip	journey, travelers, vacation, detour, route
paris	universelle, brussels, academie, montparnasse, seine
stop	refuel, restart, stops, exiting, halt
write	read, decipher, rewrite, compose, writing
language	languages, dialect, agglutinative, pidgin, lojban
beautiful	lovely, delightful, charming, exquisite, magnificent
bad	good, ugly, awful, horrible, poor
quickly	rapidly, soon, slowly, eventually, gradually
amazing	incredible, fantastic, spider, hulk, haunting

Table 4: Top neighbors (context = 15)

Query	Top Neighbors (examples)
book	books, autobiographies, published, abridged, prefaces
trip	vacation, trips, travel, destination, arriving
paris	grenoble, poissy, chapelle, universelle, nantes
stop	glottal, stops, grab, voicing, voiceless
write	read, writing, literacy, written, compile
language	languages, spoken, dialect, monolingual, vocabularies
beautiful	beauty, exquisite, lovely, quaint, magnificent
bad	good, evil, reckless, trouble, awful
quickly	swiftly, soon, eventually, rapidly
amazing	incredible, fantastic, spider, watcher, memorable

3.2 Analogy Evaluation

Table 5 summarizes representative analogy behavior. Both models produce meaningful semantic offsets (e.g., *novel:writer :: music:composer*), but some relations can be less stable depending on frequency and corpus evidence.

Table 5: Selected analogy outcomes (core evaluation)

Analogy	Ctx=5	Ctx=15
novel : writer :: music : X	composer	composer

4 Bonus Experiment 1: Combined Embeddings

4.1 Method

Skip-gram learns two embeddings per word: a target embedding ($\mathbf{v}$) and a context embedding ($\mathbf{u}$). For the bonus, we

evaluated combined vectors:

$$\mathbf{e}_{\text{combined}}(w) = \mathbf{v}_w + \mathbf{u}_w.$$

4.2 Results

Tables 6 and 7 show representative similarity neighbors using combined embeddings. Combining tends to preserve strong topical structure and can slightly “clean up” neighborhoods for some words.

Table 6: Combined embeddings: similarity (context = 5)

Query	Top Neighbors (examples)
book	books, afterword, published, preface, anthology
trip	journey, visit, vacation, route, travel, trips
paris	universelle, brussels, academie, montparnasse, seine
stop	stops, glottal, refuel, exiting, restart, halt
write	read, written, rewrite, writing, edit, compile
language	languages, agglutinative, patois, pidgin, dialect, spoken
beautiful	lovely, charming, beauty, delightful, magnificent
bad	luck, good, ugly, awful, poor
quickly	rapidly, soon, slowly, eventually, swiftly
amazing	incredible, fantastic, spider, hulk, haunting

Table 7: Combined embeddings: similarity (context = 15)

Query	Top Neighbors (examples)
book	books, scrivener, reprinted, published, abridged
trip	journey, vacation, route, visit, trips, destination
paris	modernes, bourget, neuilly, nantes, grenoble
stop	glottal, stops, grab, voicing, voiceless
write	read, writing, literacy, written, compile
language	languages, bsl, dialect, spoken, vocabularies
beautiful	beauty, exquisite, lovely, quaint, magnificent
bad	luck, good, evil, ugly, reckless
quickly	swiftly, soon, eventually, rapidly
amazing	incredible, wonder, fantastic, spider, memorable

5 Bonus Experiment 2: Gender Bias Probe

5.1 Nearest Neighbors

Table 8 compares nearest neighbors for man and woman using combined embeddings. We observe stereotypical and socially biased associations in the neighborhood of woman (e.g., marital/sexual terms), reflecting biases present in the training corpus.

Table 8: Gendered nearest neighbors (combined embeddings)

Word	Context=5 (examples)	Context=15 (examples)
man	wise, brave, madman, foolish, thief	madman, deceiver, fiend, men, shameless
woman	prostitute, pregnant, housewife, unmarried, she	womanhood, pregnant, virginity, prostitute, matrimonial

cleaner neighborhoods, and that the learned representations encode gendered stereotypes present in the corpus.

5.2 Gendered Analogy

We evaluated the analogy `man : king :: woman : X`. Table 9 reports representative outcomes. For `context=5`, `queen` appears among the top candidates (ranked within the top-15 list), whereas `context=15` returns more mixed terms (often marital/role-related) and does not surface `queen` in the top list.

Table 9: Gendered analogy: `man king woman → X` (combined)

Context	Top Candidates (examples)
5	yolande, electress, dowager, consort, <i>queen</i> (ranked)
15	dowager, matrimonial, annul, dowry, betrothed

6 Discussion

Context size. Increasing the context window from 5 to 15 substantially increased training time (Table 2) and shifted neighborhoods toward broader topical associations. This is consistent with the intuition that larger windows capture co-occurrence at a broader semantic scale, while smaller windows emphasize more local context.

Combined embeddings. Evaluating $\mathbf{v} + \mathbf{u}$ produced neighborhoods that were often similar to target-only evaluation but sometimes slightly more coherent, suggesting that target and context vectors contain complementary information.

Bias. Nearest-neighbor probes reveal stereotypical associations for gendered terms, illustrating that distributional embeddings can encode social biases present in training corpora. The analogy results further suggest that these associations can interfere with simple linear gender relations, particularly under larger context windows.

7 Conclusion

We implemented skip-gram word embeddings with negative sampling in NumPy and trained on the Wikipedia 1B corpus with context sizes 5 and 15. Both models produce meaningful similarity neighborhoods and analogical structure, with larger context sizes increasing runtime and shifting representations toward broader topical relationships. Bonus experiments show that combining target and context embeddings can yield slightly